



Development of Human-Centered, Robot-Adaptive Building Design Standards: Focusing on Korea's Building Certification Systems

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Abstract

This study aims to propose integrated design standards for human-centered, robot-adaptive buildings, focusing on circulation spaces, under the premise that service robots coexist within the existing built environment that has traditionally been designed around human users. To this end, domestic and international architectural standards related to robotics were reviewed, and a comparative analysis was conducted between the Robotics-Enabled Environment Certification (REEC), which emphasizes operational efficiency for robots, and the Barrier-Free (BF) Certification, which aims to ensure accessibility for universal users. Based on the results of this analysis, an integrated design approach was proposed by complementarily combining the requirements of the two certification systems. Through this process, design standards were derived that simultaneously secure robots' autonomous mobility and operability while maintaining human safety and accessibility. This study addresses the limitations of existing architectural standards that have been developed separately with either a human-centered or robot-centered focus. By redefining circulation spaces as areas of human-robot interaction, the research provides the baseline data to support the future institutional and practical improvement implementation of robot-adaptive architectural environments.

Keywords Robot-adaptive · Robot mobility · Smart building · Design standards · Building certification

1 Introduction

1.1 Research Background and Objectives

In recent years, our society has experienced significant socioeconomic changes—including population aging, labor shortages, infectious disease pandemics, and the rapid expansion of contactless services—which have led to a rapid increase in the utilization of service robots. In the past, robots were typically operated within spaces designated for robots. Recently, however, their deployment has expanded into human-centered living and working environments, where they actively perform various services (Lee 2021). As a result, maintaining both the safety and efficiency of operating robots inside buildings has become a

new challenge in architectural design. This shift has fueled the need for architectural spatial research focused on environments where humans and robots can coexist and work together (Lee 2024).

Humans and robots differ fundamentally in terms of physical form, environmental perception systems, mobility mechanisms, and operational modes. In particular, conventional industrial robots were manufactured as fixed systems and operated only within specially designed robot-exclusive spaces that assumed strict separation from humans. In contrast, service robots designed for collaboration and coexistence with humans are deployed within existing built environments that were originally designed for human users, requiring them to adapt to these environments (Industrial Standards Council 2024).

As a result, spatial layouts and circulation systems originally planned for human users present various obstacles for robots, effectively placing robots in a position analogous to that of users with limited mobility. This situation has led to operational inefficiencies and safety concerns, and failures of service robots deployed in real-world environments have already been documented. In Korea, an administrative

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service robot that had autonomously navigated a city hall building for nearly a year was found completely destroyed after plunging down a staircase.¹ In the United States, a sidewalk delivery robot crashed through a glass panel after its onboard sensors failed to detect the transparent surface.² These cases demonstrate that even commercially deployed robots equipped with multiple sensors remain vulnerable to ordinary features of environments designed exclusively for human users.

Against this background, the concept of robot-adaptive buildings—architectural design focused on facilitating the unobstructed mobility and smooth operation of robots—has emerged. However, concerns have also been raised that robot-centered design may marginalize human users and undermine the usability and value of architectural spaces. Accordingly, it is necessary to establish integrated design standards that are based on design principles considering universal users, including humans, while also reflecting the physical and functional characteristics of robots so that both accessibility and operational efficiency can be achieved simultaneously. Furthermore, there have been calls for the development of detailed design manuals and guidelines as well as institutional support and regulatory improvements to promote their practical implementation (Lee et al. 2022; Jung et al. 2023).

In response, this study aims to enhance the inclusiveness and sustainability of architectural spaces by integrating the concept of robot coexistence into conventional human-centered architectural design. Specifically, it seeks to develop human-centered, robot-adaptive building design standards and to provide a preliminary reference and future directions for the improvement of related legal and institutional frameworks.

1.2 Research Methods and Scope

To develop human-centered, robot-adaptive building design standards as described above, this study first investigates existing domestic and international building certification systems and then derives integrated design standards that simultaneously consider both human users and robots.

To this end, the study first explores robot-adaptive building certification systems and design standards implemented both in Korea and internationally, and identifies key issues relevant to the development of design guidelines specifically tailored to robot operation. Subsequently, a comparative analysis is conducted between the detailed design criteria of the Robotics-Enabled Environment Certification (REEC),

a certification system specialized for robot operation, and the Barrier-Free (BF) Certification, which aims to improve accessibility and convenience for universal users. Through this analysis, elements commonly pursued by both systems as well as areas where mutual complementarity is required are identified. Finally, based on the results of this analysis, the study proposes a structured, integrated design standard that incorporates specific physical specifications and performance requirements. The dimensional and performance values presented through this process are derived from the detailed criteria of the existing certification systems and standards, and the derivation logic is described in Chap. 4.

The scope of this study is limited to service robots that are similar in size to humans or smaller and that autonomously navigate within indoor built environments. Fixed industrial robots and large-scale logistics robots are excluded from the scope of this research. The spatial scope focuses primarily on circulation spaces that are essential for expanding the service range of robots. Within this framework, the analysis is limited to key planning elements that constitute such spaces—namely entrances (doors), elevators, and passageways (hallways and access routes). While the current scope is deliberately limited, directions for future research and expansion are also considered.

The following is a flowchart illustrating the research methodology and procedures of this study (Fig. 1).

2 Review of Domestic & International Standards for Robot-Adaptive Buildings

2.1 Domestic Standards for Robot-Adaptive Buildings

In Korea, various certification systems and design standards related to buildings have been established and implemented through legislation. These include the Intelligent Building System (IBS) Certification, the Green Standard for Energy and Environmental Design (G-SEED), the Zero Energy Building (ZEB) Certification, the Long-Life Housing Certification, and the Barrier-Free (BF) Certification.

However, these systems are primarily structured around accessibility, energy performance, and infrastructure environments, and they do not evaluate robot operation as a formal assessment criterion. Nevertheless, certain evaluation items within these systems have direct or indirect relevance to the operation of robotic services, particularly in areas such as technological infrastructure, physical accessibility, and spatial planning. Among them, the BF Certification is especially relevant, as it addresses spatial planning intended to ensure accessibility for universal users and therefore has

¹ SBS News. (2024, June 26). Specially hired ‘robot civil servant’ found shattered after suddenly charging down a staircase.

² SBS News. (2026, March 27). ‘Bang’ in the middle of the city... “Again after just a few days?” Anxiety spreads.

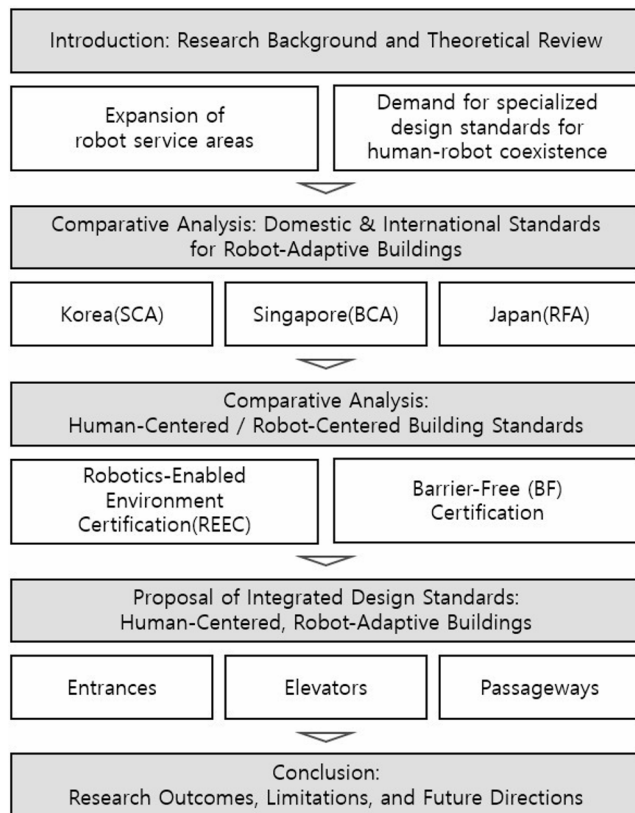


Fig. 1 Research Flowchart

a direct relationship with the implementation of robotic services within buildings.

In Korea, the certification system that directly addresses robotic operational environments is the Robotics-Enabled Environment Certification (REEC), administered by the Smart City Association (SCA). REEC is a building-level evaluation framework aimed at creating environments in which humans and robots can coexist within buildings. The certification system consists of more than 100 detailed evaluation items, covering aspects such as architectural and facility design, network and system infrastructure, building operation and management, and the implementation level of robot-support services. However, unlike the previously mentioned certification systems, REEC is a privately initiated certification scheme and does not carry legal enforceability, nor is it incorporated into the formal legal and regulatory framework.

2.2 International Standards for Robot-Adaptive Buildings

Internationally, guidelines and standards that consider robot operation within the built environment are also being developed. Representative cases can be found in Singapore and Japan.

In Singapore, the Building and Construction Authority (BCA) incorporated a section related to “Robotics & Automation” in the most recent revision of its “Design for Maintainability Guide”. This addition presents design directions that facilitate the integration of robotics and automation technologies into building maintenance activities. The section consists of 22 detailed items that systematically address aspects such as maintenance accessibility within circulation spaces, the selection of materials and finishes, equipment layout, and technological integration. However, rather than presenting specific numerical criteria, the guide mainly provides recommended design directions and considerations. Therefore, it does not function as a certification system that formally evaluates or awards robot-adaptiveness.

In Japan, the Robot Friendly Asset Promotion Association (RFA) has published the RFA Manual, a set of standard guidelines intended to support the unobstructed mobility and smooth operation of service robots. The RFA Manual establishes a structured set of criteria for the physical conditions required for service robot mobility environments. It consists of a total of 15 items related to horizontal circulation, vertical circulation, and operational environments. A notable feature of this framework is that the standards are established based on the Japanese Industrial Standards (JIS). In addition, the level of robot-friendliness within building environments is classified into three grades—A, B, and C. However, the RFA standards function as evaluation indicators voluntarily developed and operated by private stakeholders to promote the adoption of robotic services. As such, they also lack legal enforceability.

2.3 Comparative Analysis of Domestic and International Standards for Robot-Adaptive Buildings

A comparison of Korea’s REEC, Singapore’s BCA guide, and Japan’s RFA manual (Table 1) shows that all three frameworks share a common feature: they present guidelines intended to support robot operation within buildings, primarily focusing on circulation spaces. In particular, entrances, passageways, and elevator connections—representing both horizontal and vertical circulation environments—are established as key areas of consideration, with a focus on ensuring robot mobility, accessibility, and integration with building facilities and systems. This reflects the recognition that, because the functional operation of service robots presupposes spatial movement, circulation spaces constitute a critical planning element.

However, the BCA guide in Singapore functions as a set of recommended guidelines, while both the REEC and the RFA manual are privately operated certification and evaluation frameworks that lack legal enforceability. Accordingly,

Table 1 Comparison of Domestic & International Standards for Robot-Adaptive Buildings

Category	Korea	Singapore	Japan
Nature of System	Private voluntary certification system	Recommended design guideline	Private standard grading indicator
Operating Body	Private association (Smart City Association)	Public government agency (Building and Construction Authority)	Private association (Robot Friendly Asset Promotion Association)
Primary Objective	Optimization and certification of robotic operations	Enhancement of building maintenance	Standardization and grading of environments for robot deployment
Evaluation Items	102 items	22 items	15 items
Primary Evaluation Targets	Robot mobility, integration with building systems, operational support service implementation	Robot mobility, materials and finishes, integration with building systems	Robot mobility, Materials and finishes

these standards are not mandatory regulations formally incorporated into the existing architectural legal and institutional system. Rather, they rely on voluntary adoption, which limits their effectiveness from an institutional standpoint.

Furthermore, the three standards commonly adopt an efficiency-focused approach focused on robot mobility and operation. As a result, comprehensive architectural planning principles from the perspective of universal spatial use—including human users—remain relatively insufficient. While the physical movement conditions of robots and related system requirements are presented in considerable detail, design principles addressing interaction—such as circulation conflicts and safety considerations when humans and robots share the same space—have not yet been systematically established. These limitations indicate that the development of frameworks premised on human–robot coexistence remains incomplete, highlighting the need for integrated design standards that account for both human and robot activities.

3 Comparative Analysis of Human-Centered and Robot-Adaptive Standards

3.1 Comparison of Certification Systems and Evaluation Frameworks

In recent years, discussions have increasingly focused on expanding the conventional spatial planning—traditionally developed with a human-centered perspective—to accommodate the activities of service robots. In this context, the need has emerged to consider the requirements of both humans and robots simultaneously. To address this issue, this study selected two building-related certification systems in Korea for comparative analysis: the Robotics-Enabled Environment Certification (REEC), which most directly focuses on robot operation in buildings, and the Barrier-Free (BF) Certification, which aims to enhance convenience

and accessibility for universal users. The institutional characteristics and evaluation frameworks of these two systems were compared and analyzed (Table 2).

As shown in Table 2, REEC is a certification system operated by a private organization, Smart City Association, in which evaluations are conducted on a voluntary application basis without any legally mandated requirements. In contrast, BF certification is administered by the Korea Disability Policy Development Institute and is a statutory certification system based on the Act on the Guarantee of Convenience Promotion of Persons with Disabilities, Senior Citizens, Pregnant Women and Nursing Mothers. Under this law, the certification is mandatory for public buildings and for private buildings exceeding a certain size. While REEC is a voluntary certification system intended to promote market adoption, BF certification has a strong public policy orientation aimed at ensuring accessibility rights for persons with limited mobility and accessibility needs.

There are also clear differences in terms of evaluation objectives and framework. REEC focuses primarily on optimizing the autonomous mobility and task performance efficiency of robots. Its evaluation framework includes categories such as architecture and facilities, network and systems, operation and management, and robot support elements, covering not only physical spaces but also ICT infrastructure and operational systems, thereby encompassing a broad range of considerations. In contrast, BF certification aims to remove both physical and informational barriers for all users. Its evaluation categories include transitional space, interior facilities, sanitary facilities, and guidance facilities, emphasizing spatial accessibility. Its evaluation framework also primarily focuses on ensuring physical accessibility within architectural spaces, including dimensions, slopes, level differences, and guidance systems.

Accordingly, these differences are reflected in the essential elements required by each standard. REEC includes technical components directly related to robot operation, such as the performance of doors and elevators, infrastructure based on 5G and RTLS, and charging stations. In

Table 2 Comparison of Certification Standards (REEC vs. BF Certification)

Category	Robotics-Enabled Environment Certification (REEC)	Barrier-Free (BF)
Operating Organization	Smart City Association	Korea Disability Policy Development Institute
Mandatory Status	Privately operated voluntary certification	Mandatory for public sector and certain private sector
Primary Objective	Optimization of robots' autonomous mobility and task performance	Removal of physical and informational barriers for all users
Evaluation Framework	Architecture and facilities 40% Network and systems 30% Operation and management 20% Robot support 10%	Mediating facilities 22% Interior facilities 22% Sanitary facilities 25% Guidance facilities 6% Other 25%
Essential Elements	Door and elevator performance, 5G/RTLS infrastructure, docking and charging stations	Access route slope and level differences, spatial dimensions and specifications, handrails, braille signage, alarm systems
Grading System	Best / Excellent / General ($\geq 70\%$)	Best / Excellent / General ($\geq 70\%$)
Key Issues	Network dead zones, malfunction in robot-elevator interfaces	Insufficient dimensions of restrooms and ramps, excessive level differences in parking areas and access routes
Expected Impacts	Service robots operation in delivery, cleaning, and guidance	Improved accessibility for persons with limited mobility

contrast, BF certification evaluates factors centered on the safety and mobility of human users, including the width of access routes, elimination of level differences, installation of handrails and braille guidance, and ramp standards.

In terms of the grading system, both certification frameworks share the characteristic of employing score-based evaluations that classify results into three grading levels. They also share similarities in that circulation spaces are treated as key evaluation targets and both systems provide guidelines concerning the physical and functional conditions of spaces such as entrances, passageways, and elevators. This indicates that mobility environments within buildings are recognized as fundamental planning elements that support the activities of both humans and robots.

3.2 Comparison of Certification Targets and Standards

Both certification systems designate the circulation environment within buildings as a principal evaluation target; however, the focus of evaluation differs depending on the primary user—humans or robots. The functional performance of service robots within buildings presupposes moving between spaces. Since robots share the same spaces as humans while moving, circulation spaces can be regarded as areas where human-robot interaction occurs most frequently. Accordingly, in order to examine a human-centered, robot-adaptive architectural environment, it is necessary to conduct a comparative analysis of design standards centered on circulation spaces.

In this study, the principal planning elements that constitute circulation spaces were categorized into three components for analysis: entrances, elevators, and passageways. Entrances serve as the primary interface connecting interior and exterior spaces, where automatic opening and closing functions and system integration are important considerations. Elevators represent critical infrastructure that supports moving vertically between floors and incorporates system integration with building facilities. Passageways, which include hallways and access routes, function as horizontal movement spaces where physical conditions such as width, level differences, finishing materials, and safety devices have a direct impact on usability.

(1) Comparison of Entrance Design Standards

An examination of entrance (door) design standards for robot movement shows that REEC presents criteria related to clear opening width, front and rear maneuvering space, and elimination of level differences, while emphasizing system integration elements such as automatic opening functions, robot recognition markers, and integrated control with security systems. In particular, REEC includes communication (API) linkage and control functions between doors and robots as evaluation items, highlighting operational and technological integration beyond basic physical requirements.

In contrast, BF certification provides detailed specifications for physical and sensory elements intended to ensure universal accessibility for human users, including clear opening width, front and rear clearance distances,

alternatives to level differences, and tactile and auditory indicators. While both certification systems share common requirements regarding fundamental spatial conditions—such as securing adequate opening width, ensuring activity space, and minimizing level differences—REEC places greater emphasis on system integration and control functions necessary for robot operation. BF certification, on the other hand, focuses more strongly on ensuring safety and convenience for human users, including those with mobility limitations.

(2) Comparison of Elevator Design Standards

In elevator planning that considers robot usage, REEC identifies integration with standardized control systems—such as call functions, floor selection, and door control—as key elements. From an operational perspective, it evaluates aspects such as automatic robot-based call systems, synchronized door operation, and network-based communication environments. Furthermore, REEC emphasizes integrated operational systems at the facility level by including evaluation items related to the proportion of elevators configured for robot use and the degree of system connectivity.

In comparison, BF certification provides detailed regulations regarding physical dimensions and mechanical safety requirements, including the clear width of elevator entrances, interior clear floor space, and waiting space in front of the elevator. It also incorporates information accessibility elements such as braille buttons and voice guidance in order to ensure human-centered safety and ease of use. Although both systems consider elevator door operation and user safety as important factors, they differ in emphasis. REEC focuses more on integration with control systems and operational efficiency, whereas BF certification prioritizes the appropriateness of the space size and detailed dimensions.

(3) Comparison of Passageway Design Standards

A comparison of standards for passageways shows that, in consideration for situations such as passing between humans and robots, robot-to-robot encounters, waiting, and temporary occupation of space, REEC presents comprehensive criteria including hallway width, provision of waiting areas, operational hubs, electrical and communication infrastructure, and floor materials. This shows that circulation spaces are regarded not merely as moving spaces but as activity spaces where robot operations occur.

In contrast, BF certification provides pedestrian protection-oriented standards such as hallway and access route width, elimination of level differences, slip-resistant flooring performance, and installation of continuous handrails.

While both systems share the objective of ensuring safe moving through sufficient width and appropriate floor performance, their emphasis differs. REEC focuses on operational conditions that consider passing and operational scenarios, whereas BF certification emphasizes physical standards and guidance systems aimed at supporting individuals with mobility limitations and ensuring user safety.

3.3 Section Summary

A comparison between BF certification and REEC certification reveals that, although both standards address the same circulation spaces, they exhibit systematic differences in their design approaches. Both BF and REEC present criteria related to clear width, level differences, slopes, and activity space. However, BF places emphasis on securing safety and accessibility through prescriptive standards centered on physical form and dimensions, whereas REEC combines performance-based requirements—such as automated control, system integration, and communication infrastructure—to emphasize the autonomous mobility and operational efficiency of robots. Accordingly, integrated design standards should combine geometric requirements and performance-based criteria in a complementary manner, thereby ensuring both universal usability for humans and robots and operational efficiency (Table 3).

4 Proposal of integrated design standards

The integrated design standards proposed in this chapter are constructed by cross-referencing the detailed criteria of the certification systems and guidelines reviewed in Chaps. 2 and 3. The three-level structure (minimum, recommended, and optimal) takes the criteria of the statutory certification as its baseline and assigns progressively enhanced criteria derived from the comparative analysis to the recommended and optimal levels.

4.1 Human-Centered, Robot-Adaptive Entrances (Doors)

In this study, the design standards for entrances (doors) are integrated by adopting the numerical criteria and dimensional requirements of BF certification as a foundation, while incorporating the robot integration requirements of REEC standard to ensure operational functionality. For the proposed robot-adaptive entrance design, the minimum clear opening width of doors in general areas is set at 0.9 m, with 1.0 m recommended and 1.2 m or greater defined as the optimal standard. In high-traffic or congested areas, the minimum requirement is increased to 1.2 m or greater. The

Table 3 Framework for Integrating Human-Centered and Robot-Adaptive Standards

Category	BF Certification	REEC Certification	Integrated Design Framework
Primary Targets	Humans, mobility-disadvantaged users	Service robots	Humans + robots
Key Objective	Ensuring safety and accessibility	Autonomous mobility and operational efficiency	Establishing a coexistence environment
Spatial Definition	Pedestrian-oriented space	Robot operation space	Human-robot interaction space
Design Approach	Prescriptive standards based on form and dimensions	Functional standards based on performance and operation	Geometric basis + performance integration
Evaluation Criteria	Clear width, level differences, slopes, activity space	Control automation, robot integration system	Physical spatial standards + robot integration system

Table 4 Proposal of Entrance Design Standards

Item / Level	Minimum	Recommended	Optimal
Clear opening width	0.9 m	1.0 m	1.2 m (congested areas)
Access route width	1.2 m	1.5 m	1.8 m
Front and rear clear floor space	1.2 m	1.5 m	1.8 m
Threshold height difference	≤ 20 mm	0 mm	Same as the left
Access ramp slope	≤ 1/12	≤ 1/18	≤ 1/24
API-Integrated automatic doors	1 unit or more	80% or higher	100%

effective width of access routes leading to entrances must be at least 1.2 m. The required clear floor space in front of and behind the door is defined as a minimum of 1.2 m, with 1.5 m recommended and 1.8 m established as the optimal standard. Level differences at thresholds, in principle, shall be eliminated. Where unavoidable, a maximum height difference of 20 mm is permitted as the minimum requirement.

API-integrated automatic doors for robot operation must be installed at a minimum of one unit per zone where robots are expected to operate. Installation in at least 80% of all entrances is recommended, while full (100%) implementation is set as the optimal standard. Additionally, entrance doors should be equipped with robot-readable markers and secure access control systems to simultaneously ensure accessibility and operational efficiency. In areas with high traffic, crossing, or security restrictions, the installation of dedicated robot-only entrances is recommended to reduce interaction density between pedestrians and robots and to enhance safety. The following table summarizes the proposed design standards for robot-adaptive entrances (Table 4).

4.2 Human-Centered, Robot-Adaptive Elevators

The robot-adaptive elevator design standards proposed in this study adopt a phased integration approach, in which the geometric requirements of BF certification are first satisfied, followed by the incorporation of performance-based requirements from REEC standard. From a complementary perspective, the scope of API integration for communication

between robots and elevators is proposed to cover state transitions and error handling across the full operational sequence including external call, waiting, boarding, floor selection, and alighting. In addition, interior elevator signage is proposed to provide clear and legible information necessary for both humans and robots, including boarding and alighting floors, current floor location, direction of movement, and door open / close status.

The clear door opening width for elevator access is proposed as 0.8 m minimum, 1.0 m recommended, and 1.2 m as the optimal standard. The required clear floor space in front of the elevator is set at a minimum of 1.4 m × 1.4 m, with 1.5 m × 1.5 m recommended, taking into account robot types and the frequency of crossing. The interior clear floor space of the elevator is recommended to be at least 1.6 m (width) × 1.4 m (depth), with a minimum requirement of 1.6 m × 1.35 m. The gap between the elevator landing and the door threshold should be maintained at no more than 30 mm. Furthermore, there must be at least one elevator that supports standard API integration—enabling functions such as call, floor selection, door control, and re-opening detection—so that robots can operate it autonomously. It is recommended that such elevators account for at least 50% of the legally required number of elevators, with 100% implementation defined as the optimal standard. The design of the elevator lobby should ensure a clear width of at least 1.2 m to accommodate waiting and bypassing, and should include installations that secure adequate visibility. Control panels should primarily use physical buttons and incorporate multimodal interface elements such as braille signage and

Table 5 Proposal of Elevator Design Standards

Item / Level	Minimum	Recommended	Optimal
Clear door opening width	0.8 m	1 m	1.2 m
Front clear floor space	1.4 m x 1.4 m	1.5 m x 1.5 m	Same as the left
Interior clear floor space	1.6 m x 1.35 m (W x D)	1.6 m x 1.4 m (W x D)	Same as the left
Door gap	≤ 30 mm	Same as the left	Same as the left
Ev lobby clear width	≥ 1.2 m	Same as the left	Same as the left
API-Integrated elevators	1 unit or more	50% or higher (legal requirement)	100% or higher (legal requirement)

Table 6 Proposal of Passageway Design Standards

Item / Level	Minimum	Recommended	Optimal
Effective width	1.2 m	1.5 m	1.8 m
Width at crossing and waiting zones	2 m	2.2 m	2.4 m
Slope gradient	≤ 1/12	≤ 1/18	≤ 1/24
Coefficient of friction	≥ 0.4	≥ 0.75 (slope)	Same as the left
Floor grout spacing	≤ 10 mm	≤ 5 mm	Same as the left
Finished flooring	Low-reflective	Non-reflective	Same as the left

voice guidance. In addition, consideration should be given to guidance installations that indicate the elevator boarding area. The following table summarizes the proposed design standards for robot-adaptive elevators (Table 5).

4.3 Human-Centered, Robot-Adaptive Passageways (Hallways and Access Routes)

In planning passageways for robots, this study adopts a strategy that secures fundamental safety in accordance with BF certification while complementing accessibility and operational efficiency based on REEC criteria. In addition, for crossing and waiting zones, it is proposed to ensure open visibility, incorporate rounded corners, and install collision prevention devices such as kick plates and corner guards.

The effective width of passageways is defined in three levels: a minimum of 1.2 m, a recommended width of 1.5 m, and an optimal width of 1.8 m. In areas designated for crossing and waiting, the required widths are increased to 2.0 m, 2.2 m, and 2.4 m, respectively. The slope of passageways is also defined in grades, with maximum gradients of 1/12, 1/18, and 1/24. For floor performance, the coefficient of friction should be maintained at a minimum of 0.4 in level areas and at least 0.75 in sloped areas, with 0.75 or higher recommended overall. The finished flooring should, at a minimum, have low reflectivity, with non-reflective materials recommended. To prevent slipping and minimize sensor misdetection, the spacing of floor grouts should be limited to a maximum of 10 mm, with 5 mm or less recommended. The following table summarizes the proposed design standards for robot-adaptive passageways (Table 6).

5 Conclusion

5.1 Research Outcomes

This study was conducted with the aim of proposing directions for integrated human-robot design standards centered on circulation spaces, under the premise that service robots coexist within human-centered architectural environments. To this end, domestic and international standards for robot-adaptive buildings were reviewed, and a comparative analysis was carried out between robot-centered certification systems and human-centered certification systems.

First, the analysis of domestic and international cases revealed that robot-adaptive architectural standards and guidelines in Korea, Singapore, and Japan share a common feature: they consider operational environments for robots within buildings and treat circulation spaces—such as entrances, elevators, and corridors—as key planning elements. However, in all three cases, these standards are based on voluntary adoption rather than formal incorporation into legal and institutional frameworks. Moreover, since these standards primarily focus on robot mobility and operation, design principles that account for the simultaneous use of space by both humans and robots were found to be relatively insufficient.

As such, this study sought to derive standards that consider both human and robot movement by comparing the Barrier-Free (BF) certification, which aims to enhance accessibility for universal users, and the Robotics-Enabled Environment Certification (REEC), which emphasizes robot operational performance. Through this comparison, common elements and advantages of both systems were identified. While BF certification focuses on ensuring safety and accessibility for all users through prescriptive standards based on form and

dimensions, REEC emphasizes autonomous mobility and operational efficiency of robots through performance-based requirements such as automated control, system integration, and communication infrastructure. Based on these complementary differences, this study proposed integrated design standards for “human-centered, robot-adaptive buildings” by combining geometric and performance-based requirements, focusing on circulation spaces such as entrances, elevators, and passageways.

The design standards proposed in this study aim to enhance practical applicability and institutional acceptability by presenting an integrated framework grounded in design criteria commonly used in existing domestic certification systems. The proposed standards are expected to require continuous integration with various domestic and international certification systems and standards through gradual empirical validation.

5.2 Future Research Directions

This study is significant in that it analyzes the limitations of existing architectural certification systems and presents a framework capable of integrating robot operational requirements while maintaining human-centered design principles. In particular, it provides implications in robot-adaptive building design and construction by analyzing both geometric and performance-based requirements within circulation spaces and developing integrated design standards. These results may serve as a foundational reference for future discussions on the institutionalization of robot-adaptive architectural environments and the improvement of related legal and regulatory systems.

However, while this study primarily focuses on circulation spaces shared by humans and robots, future research should expand its scope to broader architectural domains, including robot-only spaces. It is expected that subsequent studies will further develop these findings into design manuals and guidelines applicable to architectural practice.

In addition, this study bears the limitation that the proposed standards were derived from a comparative analysis of existing certification criteria and related documents rather than from field experiments. To address this limitation, the proposed standards will be continuously developed and refined through follow-up research combining robot-navigation simulations and field experiments in operating buildings, through which the criteria presented in this study will be verified, calibrated, and expanded.

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Declarations

Competing Interests The authors declare no competing interests.

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